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STD: CSE SEC: SUBJECT: Computer Network Organisation

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Completed 13/10/24

Ex:No:1
2017/26.

1. Various Network commands in Linux & Windows.

Aim:

To study of various network commands used in Linux & windows.

Basic Networking commands:

1) arp -a : (terminal) ARP is short form of address resolution protocol, it will show the IP address of your computer along with the IP address and MAC address of your router.

Output:

gateway (10.0.2.2) at 52:54:00:12:35:02 [ether] on enp0s3.

2) hostname : (terminal) This is the simplest of all TCP/IP commands. It simply displays the name of your computer.

Output:

fedora.

3) ipconfig/all : (command)

This command displays detailed configuration information about your TCP/IP connection including router, gateway, DNS, DHCP and type of ethernet adapter in your system.

Output: Hostname: laptop-MHUMQBR6.
Physical address: 0A-00-27-00-00-0A.
DHCP enabled : No
Subnet mask : 255.255.255.0
IPV4 address : 192.168.56.1

(Command).

4) nbtstat -a : This command help solve problems with NetBios name resolution.

Output:

-a : Lists the remote machine's name table given its name.

-n : List local NetBIOS names;

-r : List names resolved by broadcast and via WINS.

5) netstat : netstat displays a variety of statistics (Command) about a computers active TCP/IP Connections.

Output:

Proto	Local address	foreign address	state
TCP	10.106.210.192:60268	a23-55-47-124: http.	time_wait.
TCP	10.106.210.192:63441	202:4070	Established.
TCP	127.0.0.1:7768	Laptop-MHUMQBR6: 52692	Established.

6) nslookup : (name server lookup) is a tool used to (command) perform DNS lookups in Linux.

Eg: -nslookup: www.google.com.

Output:

Default Server : Unknown.

Address : 10.106.210.238.

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7) pathping : Pathping is unique to window's and is basically a combination of ping and tracert commands. Pathping traces the route to the destination address then launches a 25 second test of each router along the way, gathering statistics on the rate of data loss along each hop.

Output :

Server : 10.106.210.238.

Address : 10.106.210.238#53.

8) ping : (Packet INternet groper) command is the best way to test connectivity between two nodes. Ping use ICMP (Internet Control Message Protocol) to communicate to each other devices.

Output :

64 bytes from fedora (fe 80::96 ff : 454d : ce50) : icmp-seq =
ttl = 64 time = 0.088 ms.

PING 24.21.4.8 (24.21.4.8) 56(84) bytes of data.

9) Route : Command is used to show / manipulate the IP routing table. It is primarily used to setup static routes to specific host or networks via an interface.

Output :

Kernel IP routing table

Destination	Gateway	Genmask	Flags	Metric	Ref	Use
default	-gateway	0.0.0.0	UG	100	0	0
10.0.2.0	0.0.0.0	255.255.255.0	U	100	0	0

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Some important Linux networking commands:

1) ip.

* Every administrator will need in daily work, from setting up new systems and assigning IPs to troubleshooting existing systems.

Eg: ip address shown

link (lookup 00:00:00:00:00:00 and
00:00:00:00:00:00)
inet (127.0.0.1.18' scope host lo).

2) if config:

It is a staple in many sysadmin's tool belt for configuring and troubleshooting networks, It has since been replaced by the ip command discussed above.

3) mtr:

* Serves as a network diagnostic and trouble shooting tool.

Eg: mtr google.com

Packets:

loss	snt	last	avg	best	Wst
0.0%	178	0.1	0.1	0.0	0.1

4) tcp dump.

Designed for capturing and displaying packets.

Eg: tcpdump -D

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5) ping.

* It is a tool that verifies IP-level connectivity to another TCP/IP computers by sending ICMP echo request messages.

Eg: ping google.com,

Student Observation:

1) Which command is used to find the reachability of a host machine from your device?

⇒ ping command is used to find the reachability of a host machine from your device.

2) Which command will we give the details of hops taken by a packet to reach its destination?

⇒ mtr, it shows route from a computer to a specified host mtr provides a lot of statics about each hop such as response time and percentage.

3) Which commands displays the ip configuration of your machine.

⇒ ip command can show address information manipulate routing plus display network various devices interfaces & funnels.

4) Which command displays the TCP port states in your machine?

⇒ net stat.

It displays variety of statistic about a computer active TCP/IP connection.

5) Write the modify the ip configuration in a Linux machine.

⇒ * Assigning IP address to interfaces ip address add 192.168.1.254/24.

* Deleting IP addresses.

ip address del 192.168.1.254/24. device enp-S03.

Ex:No:2 Ex:2 - Study of Network cables.

Aim:

To study of different types of Network cables and make a cross-wired cable and straight through cable using clamping/crimping tool.

a) Understand different types of Network cable:

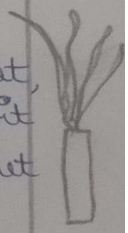
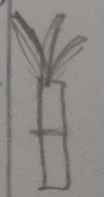
Different type of cables used in networking are

i) Unshielded twisted pair (UTP) cable,

ii) Shielded twisted Pair (STP) cable,

iii) Coaxial cable,

iv) Fibre optic cable.

Cable type	Category	Maximum data transmission	Advantages/ Disadvantages:	Use	Image
UTP	Category 3	10 bps	Advantages: * cheaper in cost. * Easy to install Disadvantages: * More prone to (EMI).	10 base-T ethernet fast ethernet, gigabit ethernet	
	Category 5	Up to 100 mbps			
	category 5e	1Gbps			
STP	Category 6, 6a	10 Gbps	Advantages: * Shielded. * Faster than UTP. * Less susceptible to noise and interference.	gigabit, ethernet, 10G ethernet (55m) widely used in data centres.	

SSTP	Category 7	10 Gbps	Disadvantages: * Expensive * Greater installation effort.	gigabit ethernet, 10G ethernet.
Coaxial cable	RG-6. RG-59 RG-11	10-100Mbps.	* High bandwidth * Low loss Bandwidth Disadvantages: * Limited distance. * Cost. * Size is bulky.	Speed of signal is 500m television network high speed internet Connections.
fibre optics cable.	Single mode Multi mode	100 Gbps	Advantages: * High speed * High Bandwidth * High Security. Disadvantages: * Expensive. * Requires skilled installers.	* Maximum distance of fibre optics cable is around 100 meters.

b) Make your own ethernet Cross - Over cable / Straight cable.

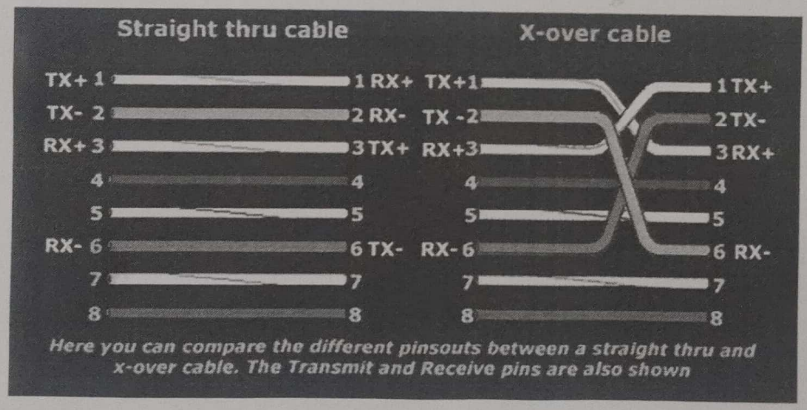
Tools and parts needed ;

* ethernet cabling. CAT 5e is certified for gigabit support, but CAT 5 cabling works as well just over shorter distances.

* A crimping tool. This is an all-in-one networking tool shaped to push down the pins in the plug and crimp and cut the shielding off the cables.

* Two RJ45 plugs,

* Optional two plug shields.



Here you can compare the different pinsouts between a straight thru and x-over cable. The Transmit and Receive pins are also shown

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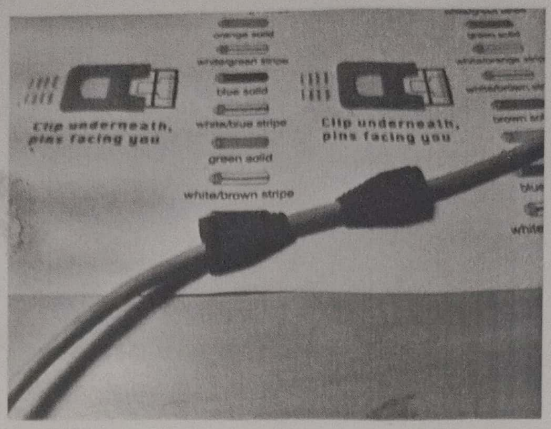
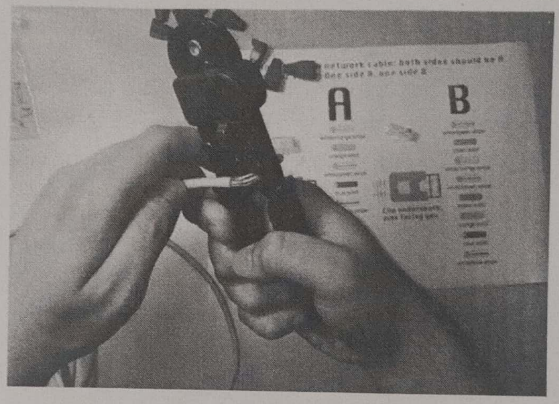
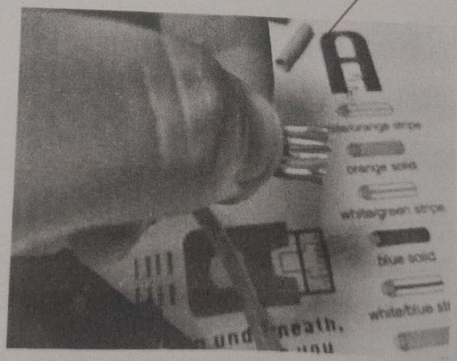
Step 1: To start construction of the device, begin by threading shields onto the cable,



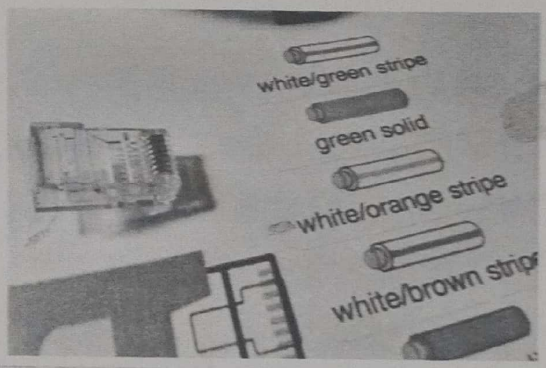
Step 2: Next, strip approximately 1,5 of cable shielding from both ends. The crimping tool has a round area to complete this task.



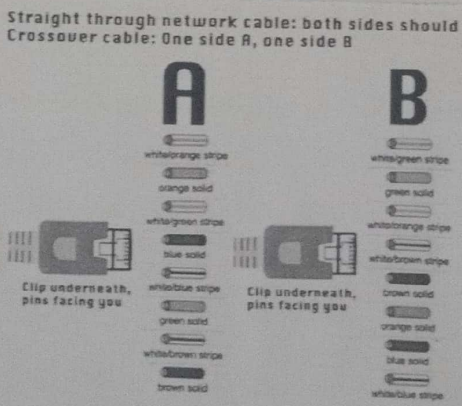
Step 3: After, you will need to untangle the wires; there should see four "twisted pairs", referencing back to the sheet, arrange them from top to bottom. One end should see in arrangement A and the other in B.



Step 4:
Once the order is correct, bunch them together in a line and if there are any that stick out farther than others, snip them back to create an even level. The difficult aspect is placing these into the RJ45 plug without messing up the order. To do so, hold the plug with the clip side facing away from you and have the gold pins facing toward you, as shown



Step 5: Next, push the cable right in. The notch at the end of the plug needs to be just over the cable shielding and if it isn't that means that you stripped off too much shielding. Simply snip the cables back a little more.



Step 6: After the wires are securely sitting inside the plug, insert it into the crimping tool and push down.

It should be shaped correctly, but pushing too hard can crack the fragile plastic plug.

Step 7: Lastly, repeat for the other end using diagram B using diagram A.

To test it, plug it in and attempt to connect two devices directly.

Student observation:

1) What is the difference between Cross cable and straight cable?

⇒

Straight cable	Cross cable.
* Same wiring standard on both ends.	* Different wiring standards on each end.
* Used to connect different types of devices.	* Used to connect similar types of devices.
* Eg: PC to switch, Router to Switch.	* Eg: PC to PC, switch to switch.

2) Which type of cable is used to connect two PCs?

⇒ Cross cable because both PCs are similar devices, so the transmit (TX) and receive (RX) wires must be crossed to communicate directly.

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3) which type of cable is used to connect a router/ switch to your PC?

⇒ Straight cable, because a PC and a switch/ router are different types of networking devices.

4) Find out the category of twisted pair cable used in your lab to connect the PC to the network socket.

⇒ Category: Cat5e or Cat6.

5) write down your understanding, challenges faced, and output received while making a twisted pair cross / straight cable.

⇒

* I learned the difference between straight and cross cables.

* I understood the T568A and T568B wiring standards.

* Arranging the wire colors in the correct order.

* Successfully prepared a straight / cross twisted pair cable.

* Tested the cable using a cable tester.

Ex: No: 3 Ex:3 - Experiments on CISCO Packet Tracer (simulation tool).

Aim:

To study the packet tracer tool installation and user interface Overview.

c) To understand environment of CISCO Packet tracer to design simple network.

Introduction:

1) It allows you to model complex systems without the need for dedicated equipment.

2) It helps you to practice your network configuration and troubleshooting skills via computer or an Android or iOS based mobile device.

3) It is available for both the linux and windows desktop environments.

4) Protocols in Packet tracker are coded to work and behave in the same way as they would on real hardware.

Installing Packet tracer.

To download Packet tracer, go to https://www.netacad.com and log in with your cisco networking academy credentials; then, click on the packet tracer graphic and download the package appropriate for your operating system.

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User interface overview:

The layout of packet tracer is divided into several components. The components of the packet tracer interface are as follows; match the numbering with explanations.

1) Menu bar - This is a common menu found in all software applications; it is used to open, save, print, change preferences, and so on.

2) Main toolbar - This bar provides shortcut icons to menu options that are commonly accessed, such as open, save, zoom, undo and redo and on the right-hand side is an icon for entering network information for the current network.

3) Logical/physical workspace tab - These tabs allow you to toggle between the logical and physical work areas.

4) Workspace - This is the area where topologies are created and simulations are displayed.

5) Common tools bar - This toolbar provides controls for manipulating topologies.

6) Real time / Simulation tabs - These tabs are used to toggle between the real and simulation modes.

7) Network component box - This component contains all of the network and end devices available with packet tracer and is further divided into two areas; Device type selection box and Device-specific selection box.

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c) User-created packet tool - users can create highly customized packets to test their topology from this area, and the results are displayed as a list.

d) Analyse the behaviour of network devices using Cisco packet tracer simulator:

i) From the network component box, click and drag and drop the below components.

a) 4 generic PCs and One hub.

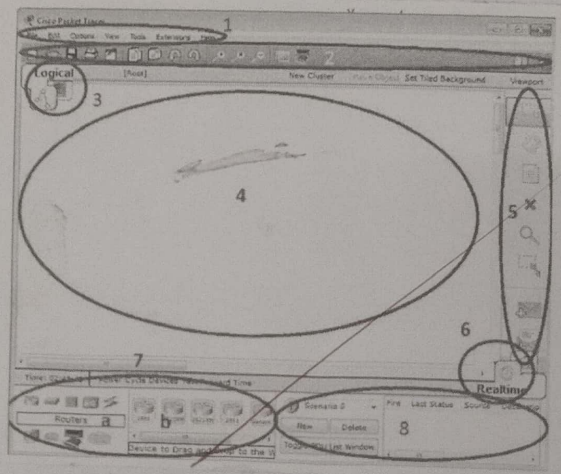
b) 4 generic PCs and one switch.

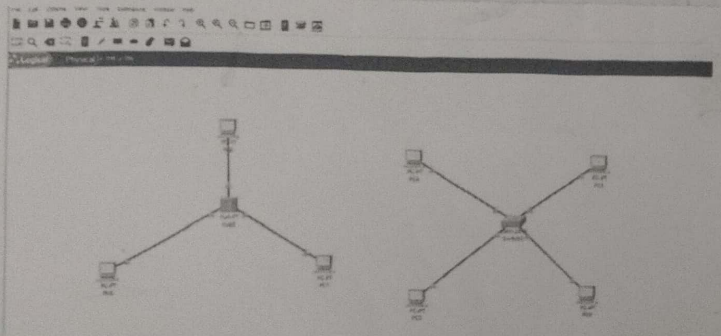
ii) Click on connections,

a) Click on copper straight-through cable.

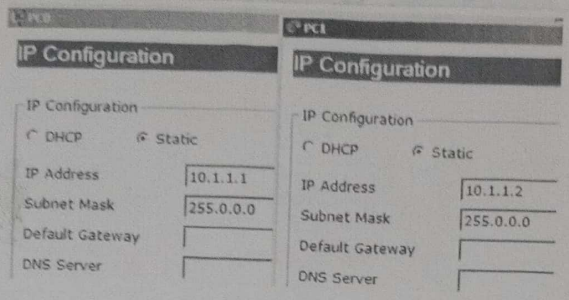
b) Select one of the PC and connect it to hub using the cable. The link LED should glow in green, indicating that the link is up. similarly connect remaining 3 PCs to the hub.

iii) Similarly connect 4 PCs to the switch using copper straight-through cable





iv) Click on the Pc's connected to hub, go to the Desktop tab, click on IP configuration and enter an IP address and subnet mask. Here the default gateway and DNS server information is not needed as there are only two end devices in the network.



v) Observe the flow of PDU from source PC to destination PC by selecting the realtime mode of simulation.

vi) Repeat step 3 to step 5 for the Pc's connected to the switch.

vii) Observe how hub and switch are forwarding the PDU and write your observation and conclusion about the behaviors of switch and hub.

Student observation:

a) From your observation write down the behaviour of switch and hub in terms of forwarding the packets received by them.

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⇒ Hub:

* A hub broadcasts the received packet to all ports connected to it, except the port from which the packet was received.

Switch:

* A switch checks the MAC address of the destination device and forwards the packet only to the specific port where the device is connected.

Device	Packet forwarding Behaviour.
Hub	Broadcast packets to all connected ports.
Switch	forwards packets only to the destination port.

b) Find out the network topology implemented in your college and draw and label that topology in your observation book.

⇒ Most modern college computer networks use a star topology, often extended into a hierarchical/ extended star topology.

Labels to include:

* Internet.
* Access switch
* Network cables.
* Core switch.

Observation:

The college network follows an extended star topology, where multiple computers are connected to access switches and the access switches are connected to a central switch.

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Ex.No:4 Setup and Configure a LAN using a switch and ethernet cables in your lab.

Aim:

To design, set-up and configure a local area network connecting four computers through a network switch, assign static IP addresses to all devices, verify connectivity using the ping command, and demonstrate resource/data sharing across the network.

Equipment required:

Equipment	Quantity.	specification
PC	4	Ethernet (NIC) enabled.
Network switch	1	8/16/24 port, unmanged / managed.
Ethernet cables (Cat 5e / cat 6)	4	Straight-through, RJ-45 connectors.

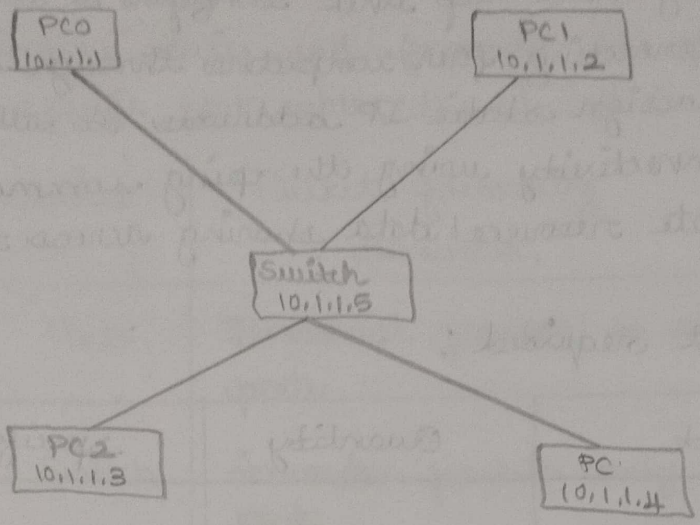
Theory:

A local area network is a network that connects computers and devices within a limited geographical area, allowing them to share resources such as files, and internet connections.

A star topology is used in this experiment, where all end devices (PCs) are individually connected to a central network switch. The switch operates at layer 2 of the OSI model and forwards data frames between connected devices on the MAC addresses. Every device on a LAN must be assigned an IP address within the same network range.

Network diagram

The LAN was implemented using a star topology as shown below, with 4 PC's connected individually to a central switch.



IP address configuration table

Device	IP address	subnet mask	Connected to
PC0	10.1.1.1	255.0.0.0	Switch port Fa0/1.
PC1	10.1.1.2	255.0.0.0	switch port Fa0/2.
PC2	10.1.1.3	255.0.0.0	switch port Fa0/3.
PC3	10.1.1.4	255.0.0.0	switch port Fa0/4.
switch	10.1.1.5	255.0.0.0.	VLAN 1 Interface.

Implementation in Cisco Packet Tracer:

Step 1; Place the devices.

1) Open Cisco Packet tracer.

2) From the device category panel, select End devices and drag 4 PCs onto the workspace.

3) Switch and drag 1 switch onto the workspace.

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Step 2: Cable the devices.

4. click connections → select copper straight-through cable.

5. click each PC, then click the switch, and select an available fast ethernet port to complete each connection.

6. Repeat for all four PCs. green link lights confirm a successful physical connection.

Step 3: Assign IP addresses to PCs.

7. click a PC → desktop tab → IP configuration.

8. Enter the static IP address and subnet mask as per the IP table for each PC.

Step 4: Configure the switch management IP.

9. click the switch → CLI tab.

10. Enter the following commands to assign a management IP:

```
enable
configure terminal
interface vlan1
ip address 10.1.1.5 255.0.0.0
no shutdown
exit
```

Step 5: Verify connectivity using ping

11. click a PC → desktop tab → command prompt.

12. Use the ping command to test connectivity to the other PCs.

13. Successful "Reply from...." messages confirm Connectivity.

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Step 6: Demonstrate data sharing (Server simulation)

14) Add a server device (End devices → Server-PT) and connect it to the switch.

15) Assign it an IP address, subnet mask via its Desktop → IP configuration tab.

16) Open the server's services tab and enable the http service.

17) From another PC, open the web browser and enter the server's IP address to load the page, confirming successful data transfer across the LAN.

Step 7: Observe packet flow in simulation mode.

18) Switch to simulation mode.

19) Send a ping or web request between two devices and use the play/capture controls to step through the packet as it travels hop-by-hop through the switch, observing MAC address based forwarding.

Observation:

All four PCs were successfully connected to the switch in a star topology and assigned unique static IP addresses within the same subnet. Ping tests between all devices returned successful replies, confirming end-to-end connectivity. The shared folder/server resource was successfully accessed from other computer on the network, confirming that the LAN was functioning correctly for resource sharing.

Ex.No:5 Experiment on Packet capture tool: Wireshark.

Aim:

To determine the packet capture tool like wireshark.

Packet Sniffer:

* Sniffs messages being sent/received from/by your computer.

* Passive program,

⇒ never sends packet itself.

⇒ no packets addressed to it.

⇒ receives a copy of all packets.

Description:

Wireshark:

* Wireshark, a network analysis tool formerly known as ethereal, captures packets in real time and display them in human readable format. Wireshark includes filters, color coding and other features that let you dig deep into network traffic and inspect individual packets.

Wireshark used for,

⇒ Network administrators: troubleshoot network problems.

⇒ Network security engineers: examine security problems.

⇒ Developers: debug protocol implementations.

⇒ People: learn network protocol internals.

Getting Wireshark:

Wireshark can be downloaded for windows or macOS from its official website. For linux or another UNIX-like system, wireshark will see found in its package repositories.

Capturing packets:

After downloading and installing wireshark, launch it and double-click the name of a network interface under capture to start capturing packets on that interface. As soon as click the interface's name, you'll see the packets start to appear in real time.

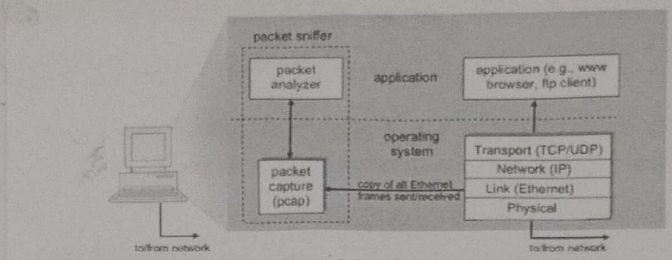
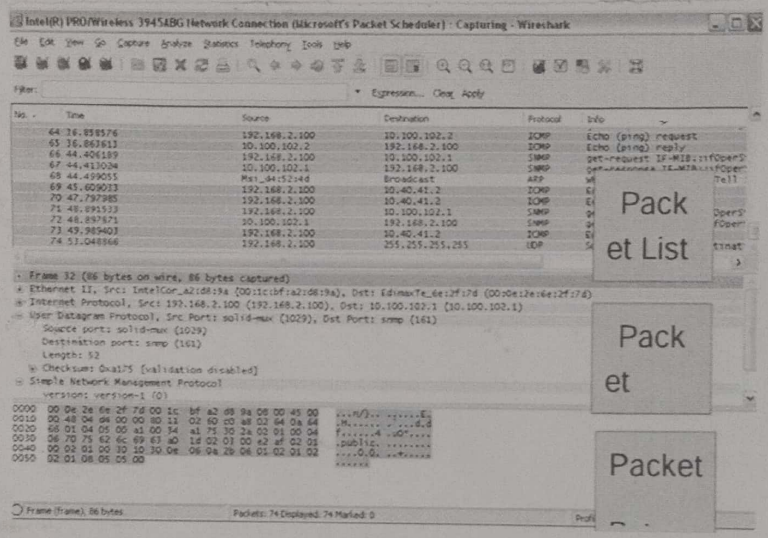


Figure 1: Packet sniffer structure



The "Packet List" Pane:

The packet list pane displays all the packets in the current capture file. "The Packet list" plane each line in the packet list corresponds to one packet in the capture file.

The "Packet details" pane:

The packet details pane shows the current packet in a more detailed form. The pane shows the protocol and protocol fields of the packet selected in the "Packet List" pane.

The "Packet bytes" Pane:

The packet bytes pane shows the data of the current packet in a hexdump style.

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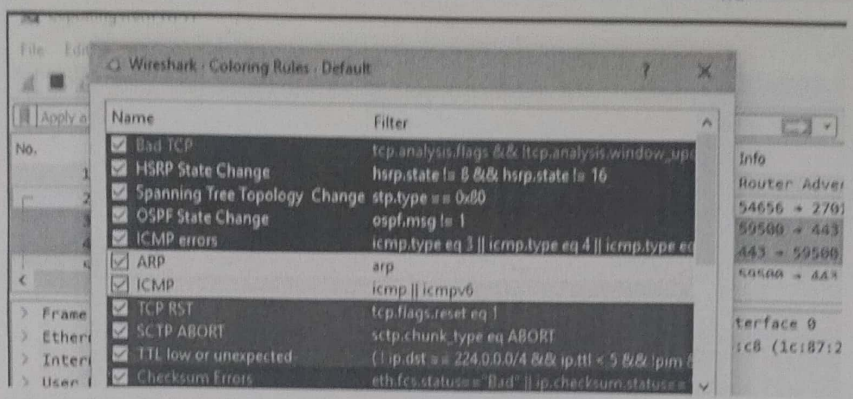
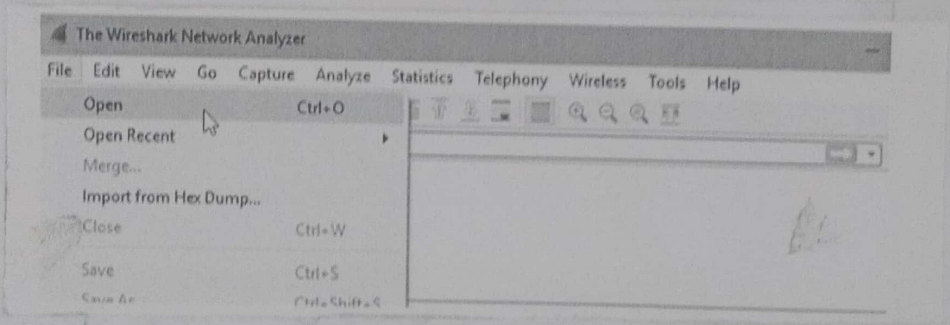
Color coding:

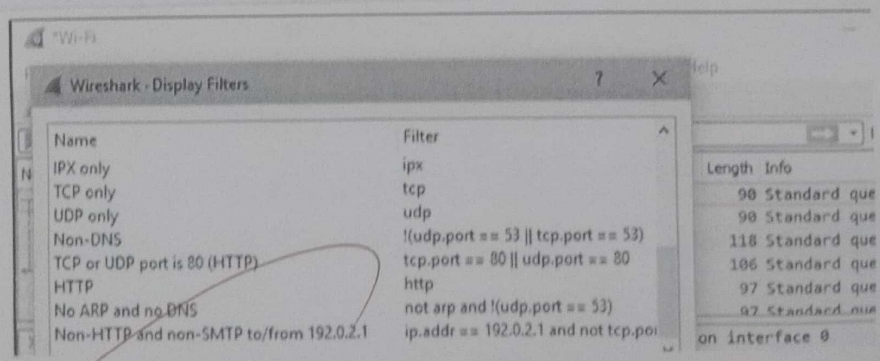
You will probably see packets highlighted in a variety of different colors. Wireshark uses colors to help you identify the types of traffic at a glance. To view exactly what the color codes mean, click View > Coloring Rules.

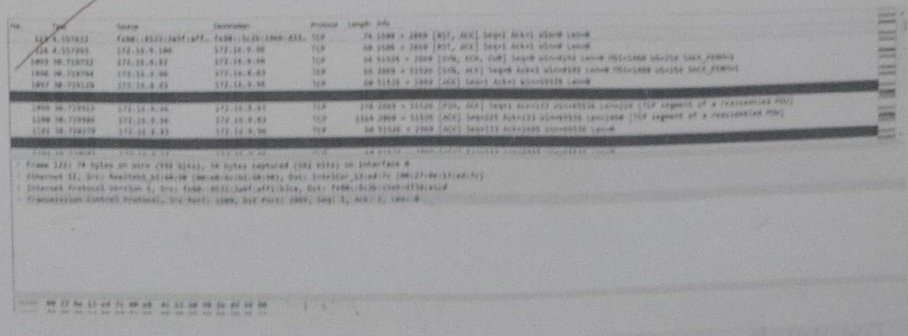
Filtering packets:

If you're trying to inspect something specific, such as the traffic a program sends when phoning home, it helps to close down all other applications using the network so you can narrow down the traffic.









Capturing and analysing packets using wireshark tool.

To filter, capture, view, packets in wireshark tool.

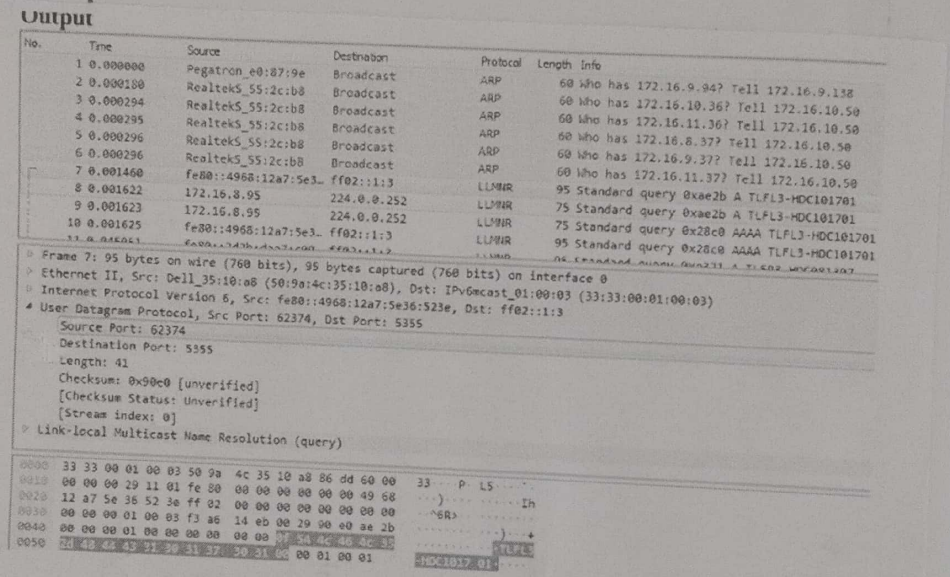
Capture 100 packets from the ethernet: IEEE 802.3 LAN interface and save it.

Procedure:

* Select local area connection in wireshark.
* Go to capture option.
* Select stop capture automatically after 100 packets
* Then click start capture.
* Save the packets.

Output:

Output



i) Create a filter to display only TCP/UDP packets, inspe the packets and provide the flow graph.

ii) Create a filter to display only ARP packets and inspect the packets.

iii) Create a filter to display only DNS packets and provide the flow graph.

iv) Create a filter to display only HTTP packets and inspect the packets.

v) Create a filter to display only IP/ICMP packets and inspect the packets.

vi) Create a filter to display only DHCP packets and inspect the packets.

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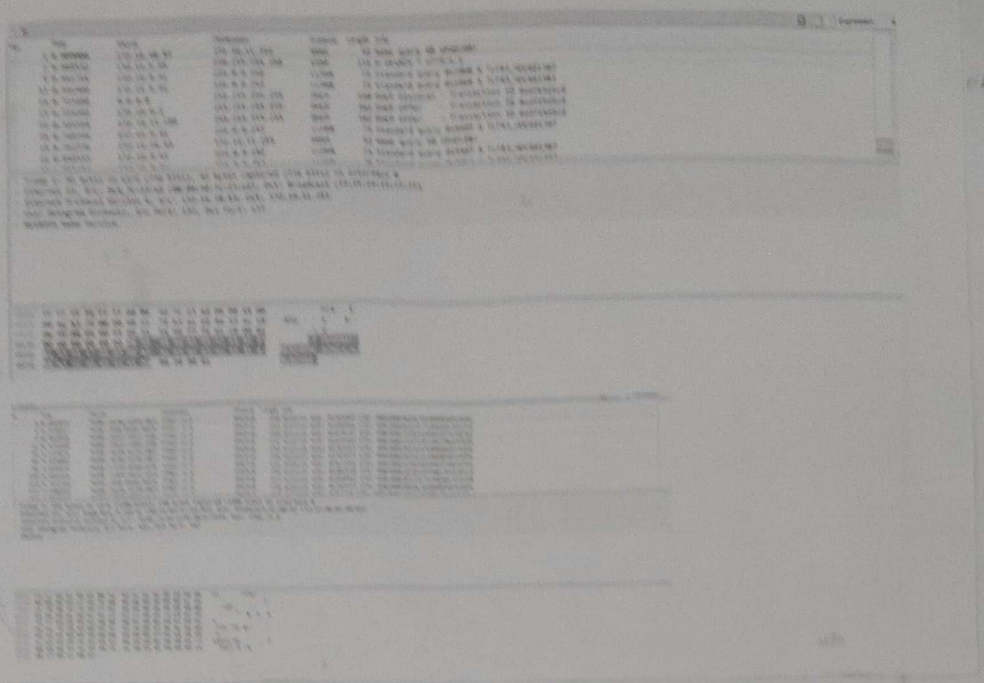

Student observation:

1) What is promiscuous mode?

⇒ It is a network interface mode in which a device captures and receives all network packets.

2) Does ARP packet have a transport layer header? Explain

⇒ No, ARP works directly at the network access and is not carried using TCP or UDP.

3) Which transport layer protocol is used by DNS?

⇒ DNS primarily uses UDP at port 53 for normal queries. It can also use TCP port 53, especially for zone transfers.

4) What is the port number used by HTTP protocol?

⇒ The port number for HTTP is 80.

5) What is a broadcast IP address?

⇒ A broadcast IP address is an IP address used to send a packet to all devices on a particular network/subnet.

Ex:No:6 Error correction at Data link layer.

Aim:

Write a program to implement error detection and correction using hamming code concept. Make a test run to input data stream and verify error correction.

Error correction at Data link layer:

* Hamming code is a set of error-correction codes that can be used to detect and correct the errors that can occur when the data is transmitted from the sender to the receiver.

Create a sender program:

⇒ Input to sender file should be a text of any length. Program should convert the text to binary.

⇒ Apply hamming code concept on the binary data and add redundant bits to it.

⇒ Save this output in a file called channel.

Create a receiver program:

⇒ Receiver program should read the input from channel file.

⇒ Apply hamming code on a binary data to check for errors.

⇒ If there is an error, display the position of the error.

⇒ Else remove the redundant bits and convert the binary data to ascii and display the Output.

Scanned with OKEN Scanner

Code : (sender).

```
#include <stdio.h>
#include <string.h>
void hanning_code (char data, char code[13])
{
  int d[9];
  int p1, p2, p4, p8;
  for (int i=0; i<8; i++){
    d[i+1] = (data >> (7-i)) & 1;
  }
  code[3] = d[1];
  code[5] = d[2];
  code[6] = d[3];
  code[7] = d[4];
  code[9] = d[5];
  code[10] = d[6];
  code[11] = d[7];
  code[12] = d[8];
  p1 = code[3] ^ code[5] ^ code[7] ^ code[9] ^ code[11];
  p2 = code[3] ^ code[6] ^ code[7] ^ code[10] ^ code[11];
  p4 = code[5] ^ code[6] ^ code[7] ^ code[12];
  p8 = code[9] ^ code[10] ^ code[11] ^ code[12];
}
int main(){
  file *fp;
  char text[1000];
  char code[13];
  printf("Enter the text: ");
  fgets(text, sizeof(text), stdin);
  fp = fopen("channel.txt", "w");
  if (fp == NULL)
  {
    printf("Error opening channel");
    return 1;
  }
```

```
for(int i=0; i<strlen(text); i++) {
    if(text[i] == '\n')
      break;
    hamming_encode(text[i], code);
    for(int j=1; j<=12; j++) {
      fprintf(fp, "%d", code[j]);
    fprintf(fp, " ");
  }
  fclose(fp);
  printf("Data encoded successfully");
  return 0;
}
```

Code : (receiver).

```
#include <stdio.h>
int main(){
    file *fp;
    char block[20];
    int code[13];
    int data;
    char ch;
    int error_position;
    fp=fopen("channel.txt","r");
    if(fp==NULL){
    printf("Error");
    return 1;
    }
    printf("\n Received data:\n");
    while(fscanf(fp,"%s",block)==1){
      for(int i=1; i<=12; i++){
          code[i]=block[i-1]-'0';
      }
      error_position = p1*1 + p2*2 + p4*4 + p8*8;
```

```
if (error_position == 0){
  printf ("No errors detected");
}
else {
  code [error_position] = code [error_position] ^ 1;
  printf ("Error corrected");
}
data = 0;
data = (data << 1) | code [3];
data = (data << 1) | code [5];
data = (data << 1) | code [6];
data = (data << 1) | code [7];
data = (data << 1) | code [9];
data = (data << 1) | code [10];
data = (data << 1) | code [11];
data = (data << 1) | code [12];
  ch = (char) data;
  printf (" Received character: %c \n", ch);
}
  fclose (fp);
  printf ("Data received successfully. \n");
  return 0;
}
```

Output:

Enter the text: Hello

Data encoded successfully.

Hamming Coded data is stored in channel.txt.

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EX:NO:7 Flow control at Data link layer.

Aim:

Write a program to implement flow control at data link layer using sliding window protocol.

Create a sender program:

i) Input window size from the user.

ii) Input a text message from the user.

iii) Consider 1 character per frame.

iv) Create a frame with following fields.

v) Send the frames.

vi) Wait for the acknowledgement from the receiver.

vii) Reader a file called receiver-buffer.

viii) Check ACK field for the acknowledgement number.

Create a receive file:

i) Reader a file called sender-buffer.

ii) check the frame no.

iii) If the frame no are as expected, write the approxiate ACK no in the receiver-buffer file.

code;

Sender program;

```
#include <stdio.h>
int main() {
  int window, i, start = 0, ack;
  char msg[100];
  file* fp;
  printf("Enter window size: ");
  scanf("%d", &window);
  int n = strlen(msg);
  while (start < n) {
    fp = fopen("Sender-buffer.txt", "w");
    printf("Sending frames ":);
    for(int i = start; i < start + window && i < n; i++) {
      printf("Frame No: %d, Data: %c\n", i+1, msg[i]);
    }
    fclose(fp);
    fp = fopen("Receiver-buffer.txt", "r");
    if (fp == NULL) {
    printf("Receiver-buffer.txt not found\n");
    continue;
    }
    fscanf(fp, "%d", &ack);
    fclose(fp);
    if (ack == start + window || ack >= n) {
      printf("ACK %d received", ack);
      start += window;
    }
    else {
      printf("Resending frames from frame %d:\n",
                                    start+1);
      }
    }
    return 0;
}
```

Receiver program:

```
#include <stdio.h>
#include <unistd.h>
int main() {
    FILE *fp, *ackfp;
    char line[100];
    int frame, expected = 1;
    char data;
    while (1) {
     fp = fopen ("sender_buffer", "r");
     if (fp==NULL)
    {
     printf ("Waiting for sender");
      sleep (2);
      continue;
    }
    while (fgets (line, sizeof(line), fp)) {
    ackfp = fopen ("Receiver buffer", "w");
     if (frame == expected) {
       printf ("Frame %d accepted\n", frame);
        expected ++;
     }
    else {
        printf ("Frame %d rejected", frame);
        }
        fclose (ackfp);
      }
     fclose (fp);
     sleep (3);
  }
   return 0;
  }
```

Ex.No:8 NMAP to Discover live hosts using Nmap scans on the TryHackMe Platform.

Aim:

To discover live hosts using Nmap scans (ARP, ICMP, TCP/UDP) on the TryHackMe platform Room.

Introduction:

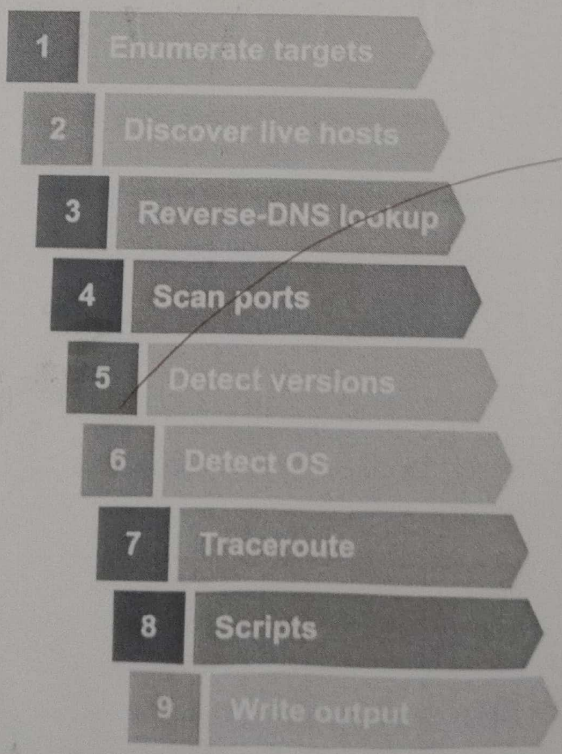
This room is the first of four in this Nmap series. These four rooms are also part of the Network Security module.

⇒ Nmap live host discovery.
⇒ Nmap basic port scans.
⇒ Nmap advanced Port scans.
⇒ NMAP post port scans.

ARP scan: This scan uses ARP requests to discover live hosts.

ICMP scan: This scan uses ICMP requests to identify live hosts.

TCP/UDP ping scan: This scan sends packets to TCP ports and UDP ports to determine live hosts:



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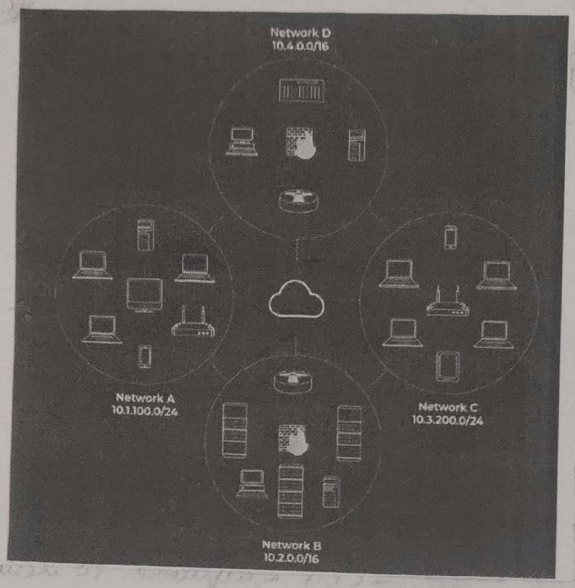
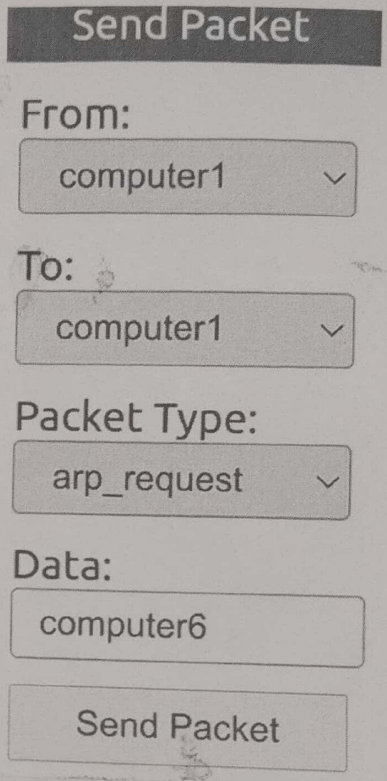
Subnetworks:

A network segment is a group of computers, linked through a shared medium, like an ethernet switch or WiFi access point. In IP networks, a subnetwork typically consists of one or more network segments connected together and configured to use the same router. A network segment is a physical connection, while a subnetwork is a logical connection.



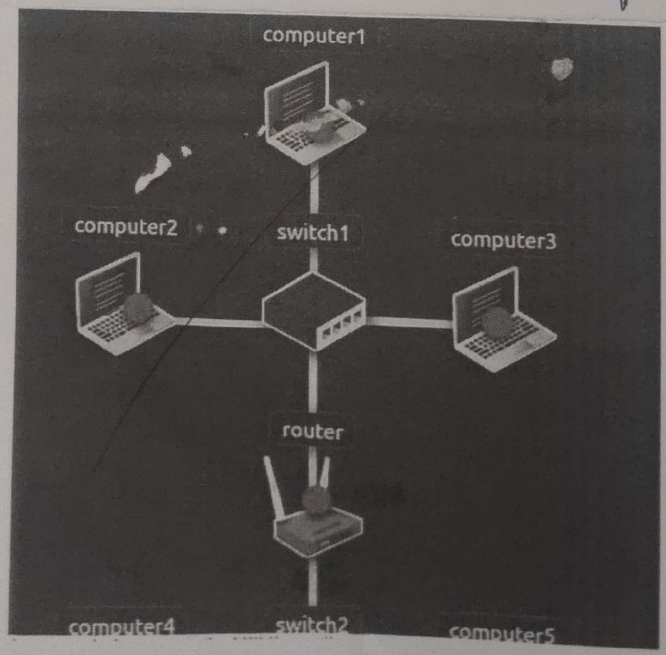


i) How many devices can see the ARP request?

⇒ 4.

ii) Did computer 6 receive the ARP request? (Y/N).

⇒ N.



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i) How many devices can see the ARP request?

⇒ 4

ii) Did computer 6 reply to the ARP request? (Y/N)

⇒ Y.

Discovering Live hosts:

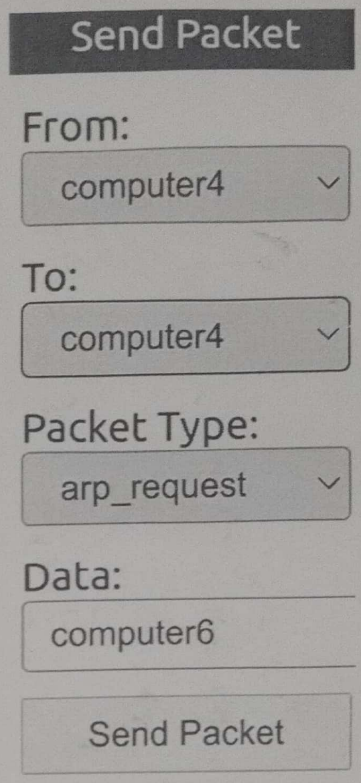
TCP/IP layers:

⇒ ARP from link layer,

⇒ ICMP from Network layer.

⇒ TCP from Transport layer.

⇒ UDP from Transport layer



What is the name of the first device that responded to the first ARP request?

⇒ router,

What is the name of the first device that responded to the second ARP request?

⇒ Computer 5,

Send another ping request. Did it require new ARP requests? (Y/N).

⇒ N.

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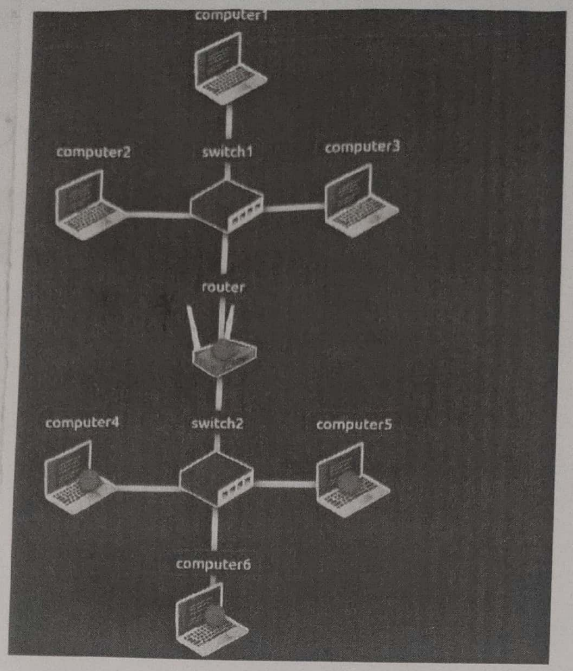

7	Application Layer	Application Layer	HTTP, HTTPS, SMTP, POP3, IMAP, SSH, FTP, SNMP, Telnet, RDP,...
6	Presentation Layer		
5	Session Layer		
4	Transport Layer	Transport Layer	TCP, UDP
3	Network Layer	Network Layer	IPv4, IPv6, ICMP, IPsec
2	Data Link Layer	Link Layer	ARP, Ethernet (802.3), WiFi (802.11), DSL, Bluetooth,
1	Physical Layer		

Pentester Terminal

```
pentester@TryHackMe$ sudo nmap -PR -sn 10.10.210.6/24

Starting Nmap 7.60 ( https://nmap.org ) at 2021-09-02 07:12 BST
Nmap scan report for ip-10-10-210-75.eu-west-1.compute.internal (10.10.210.75)
Host is up (0.00013s latency).
MAC Address: 02:83:75:3A:F2:89 (Unknown)
Nmap scan report for ip-10-10-210-100.eu-west-1.compute.internal (10.10.210.100)
Host is up (-0.10s latency).
MAC Address: 02:63:D0:10:2D:CD (Unknown)
Nmap scan report for ip-10-10-210-165.eu-west-1.compute.internal (10.10.210.165)
Host is up (0.00025s latency).
MAC Address: 02:59:79:4F:17:67 (Unknown)
Nmap scan report for ip-10-10-210-6.eu-west-1.compute.internal (10.10.210.6)
Host is up.
Nmap done: 256 IP addresses (4 hosts up) scanned in 3.12 seconds
```

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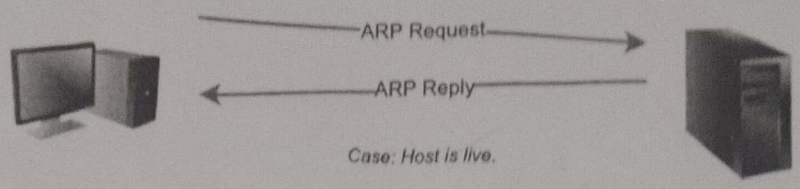
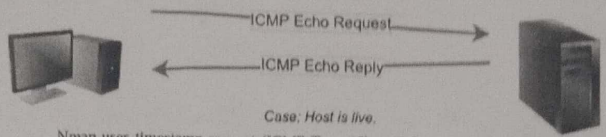
```
nmap -PR -sn TARGET
```



Case: Host is live.

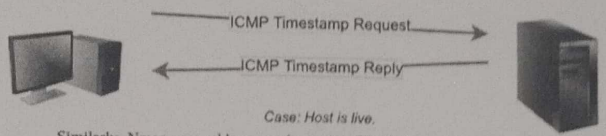
```
nmap -PE -sn TARGET
```



Case: Host is live.

Nmap uses timestamp request (ICMP Type 13) and checks whether it will get a Timestamp reply (ICMP Type 14). Adding the -PP option tells Nmap to use ICMP timestamp requests. As shown in the figure below, you expect live hosts to reply.

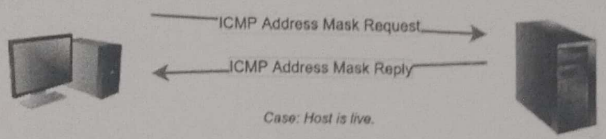
```
nmap -PP -sn TARGET
```



Case: Host is live.

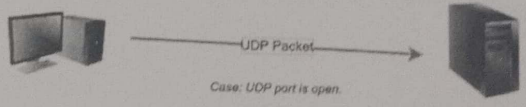
Similarly, Nmap uses address mask queries (ICMP Type 17) and checks whether it gets an address mask reply (ICMP Type 18). This scan can be enabled with the option -PM. As shown in the figure below, live hosts are expected to reply to ICMP address mask requests.

```
nmap -PM -sn TARGET
```



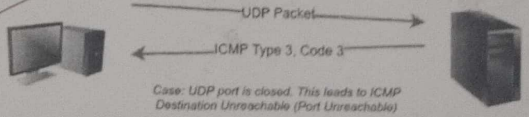
Case: Host is live.

```
nmap -PU -sn TARGET
```



Case: UDP port is open.

```
nmap -PU -sn TARGET
```



Case: UDP port is closed. This leads to ICMP Destination Unreachable (Port Unreachable)

Answer the questions below

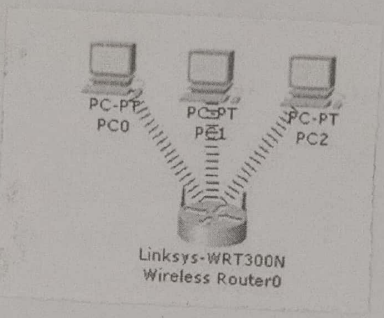
Which TCP ping scan does not require a privileged account?

Ex.No: 8b Configuration of wireless LAN using CISCO packet Tracer.

Aim:

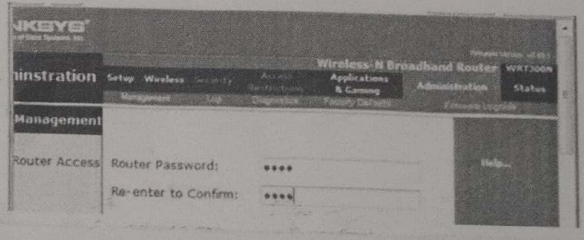
Configuration of wireless LAN using CISCO packet tracer.

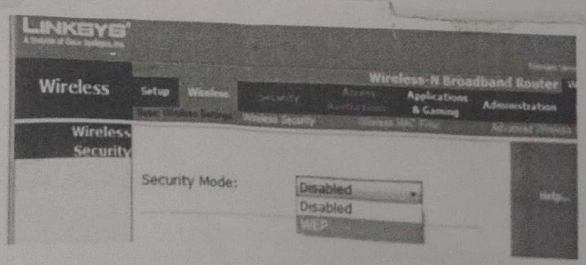
Design a topology with 3 Pc's connected from linksys wireless routers.



Perform following configuration:

* Configure Static IP on PC and wireless router.
* Set SSID to mother network.
* Set IP address of router to 192.168.0.1, PC0 to 192.168.02, PC1 to 192.168.0.3 and PC2 to 192.168.0.4
* Secure your network by configuring WAP key on router.
* Connect Pc by using WAP key.





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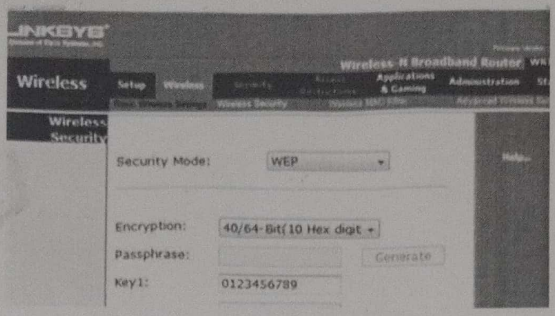
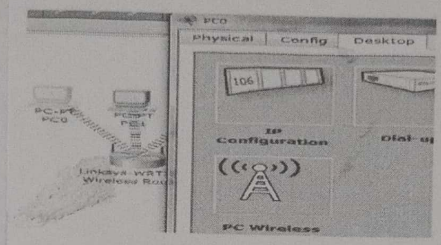
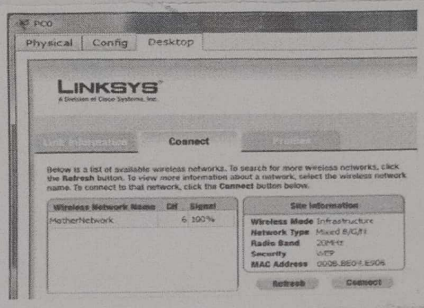
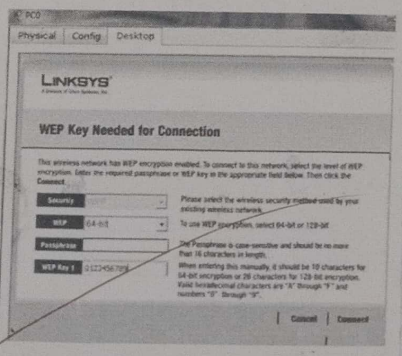
Key1 to 0123456789





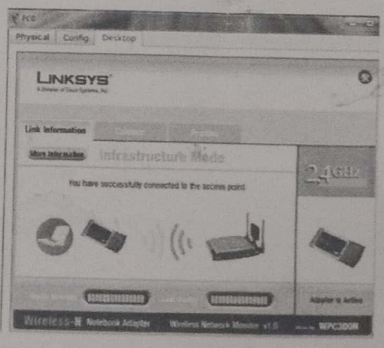
lick on connect tab and click on Refresh button





connect you with wireless router.

u can see in image below that system is connected. And PCI card is



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Student Observation:

c) What is SSID of a wireless router?

⇒ It is the name of a wireless network / router used to identify and connect to a Wi-Fi network.

d) What is the security key in a wireless router?

⇒ It is the password used to authenticate and secure a wireless network.

e) Configure a simple wireless LAN using a real access point?

⇒ i) Connect the access point to the computer.

ii) Open the router's configuration page.

iii) Set the SSID.

iv) Select a security method such as WAP2 / WPA3.

v) Set a password.

vi) Save the configuration & restart the router if required.

vii) Test connectivity using ping between the connected PCs.

x:NO:9 Implementation of Subnetting in CISCO packet Tracer.

Aim:

Implementation of subnetting in CISCO packet tracer simulator.

Creating a network topology:

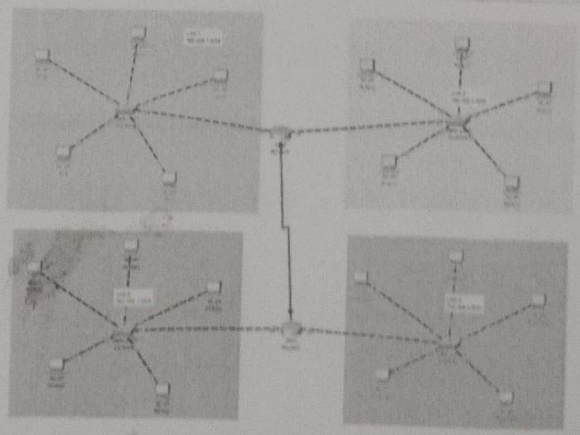
The first step in implementing classless IP subnetting is to create a network topology in packet tracer. To create a network topology in Packet tracer, select the "New" Button in the top left corner, then select "Network" and "generic".

Adding the Devices:

Once we have created our network topology, we can add devices to it. Here, we will be adding routers, switches and PC's. To add a device, select the device from the bottom left corner and drag it onto the network topology.

Subnetting:

To subnet the network address of 192.168.1.0/24 to provide enough space for at least 5 addresses for end devices, the switch and the router, we can use a /27 subnet mask.



The IP addressing for the network shown in the topology can be as follows:

Router R_1

Gigabit Ethernet 0/0: 192.168.1.1

Gigabit Ethernet 0/1: 192.168.2.1

Switch S_1:

Fast ethernet 0/1: 192.168.1.0/27

PC1: 192.168.1.11

PC2: 192.168.1.12

PC3: 192.168.1.13

PC4: 192.168.1.14

PC5: 192.168.1.15

Fast Ethernet 0/2: 192.168.2.0/27

PC1: 192.168.2.11

PC2: 192.168.2.12

PC3: 192.168.2.13

PC4: 192.168.2.14

PC5: 192.168.2.15

Router R_2:

Fast Ethernet 0/0: 192.168.3.1

Fast Ethernet 0/1: 192.168.4.1

Switch S_2:

Fast Ethernet 0/1: 192.168.3.0/27

PC1: 192.168.3.11

PC2: 192.168.3.12

PC3: 192.168.3.13

PC4: 192.168.3.14

PC5: 192.168.3.15

Configuring the devices:

Now that we have added our devices and connected them, we can start configuring them. We will start by configuring the router. Right-click on the router and select "CLI". This will open the command line interface (CLI) for the router.

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Testing the network:

Now that our network topology is configured, we can test the network; open a command prompt on each PC and try to ping the outer PC. If the ping is successful, then the network is functioning properly.

Student Observation:

a) Write down your understanding of subnetting.

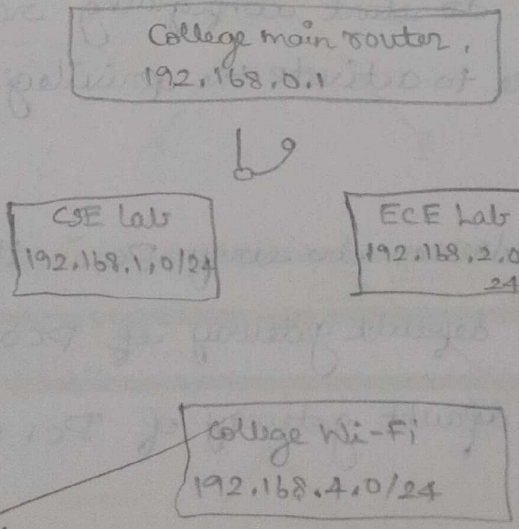
⇒ Subnetting is the process of dividing a large network into smaller logical networks called ~~network~~ subnets.

b) What are the advantages of implementing subnetting within a network?

⇒
* Reduces network traffic.
* Improves Security.
* Provides scalability.
* Simplifies network management.

c) Find out whether subnetting is implemented in your college. If yes, draw and list down the subnets used with IP addresses.

⇒ Yes, subnetting is commonly implemented in college networks to separate departments.



Conclusion:

Subnetting helps a college manage its network efficiently by dividing it into smaller networks.

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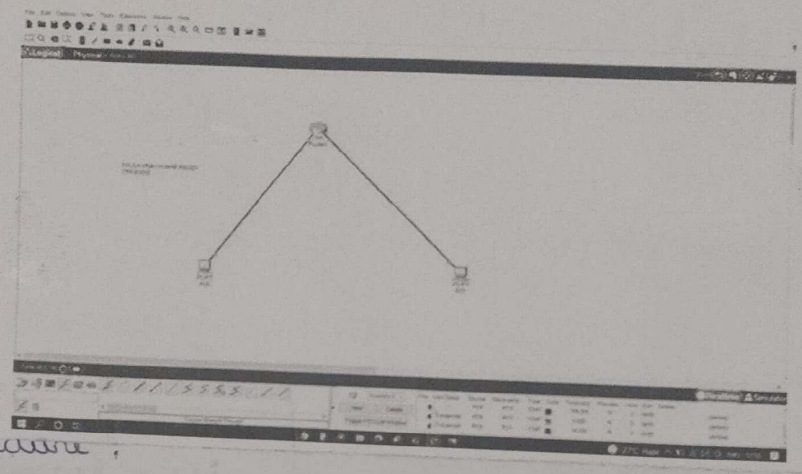
Ex:No:10 Internetworking with routers in CISCO packet Tracer Simulator

a) Aim:

Internetworking with routers in CISCO Packet tracer simulator.

d) Design and configure a simple network using a router.

In this network, a router and 2 PCs are used. Computers are connected with routers using a Copper straight-through cable. After forming the network, to check network connectivity a simple PDU is transferred from PC0 to PC1.



Procedure:

Step 1: (Configuring router 1):

i) Select the router and Open CLI.

ii) Press Enter to start configuring router 1.

iii) Type enable to activate the privileged mode.

Step 2:

i) Assign IP addresses to every PC in the network.

ii) Assign the default gateway of PC0 as 192.168.10.1

iii) Assign the default gateway of PC1 as 192.168.20.1

Step 3:

i) Connect fast ethernet 0 port of PC0 with fast ethernet 0/0 port of router 1 using a copper straight through cable.

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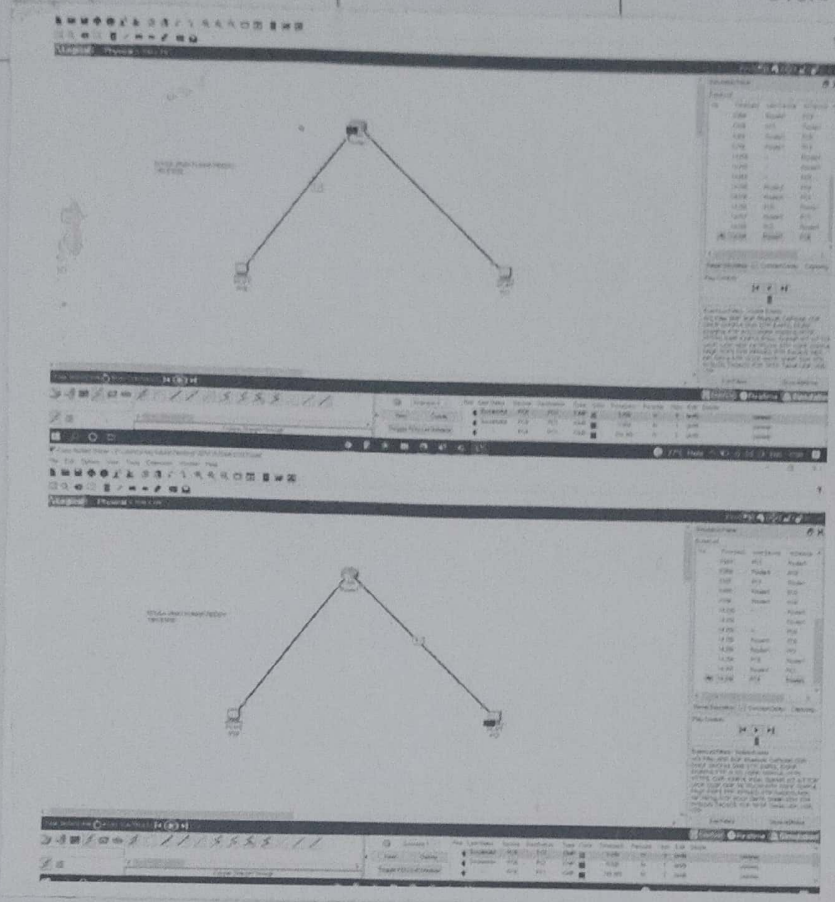
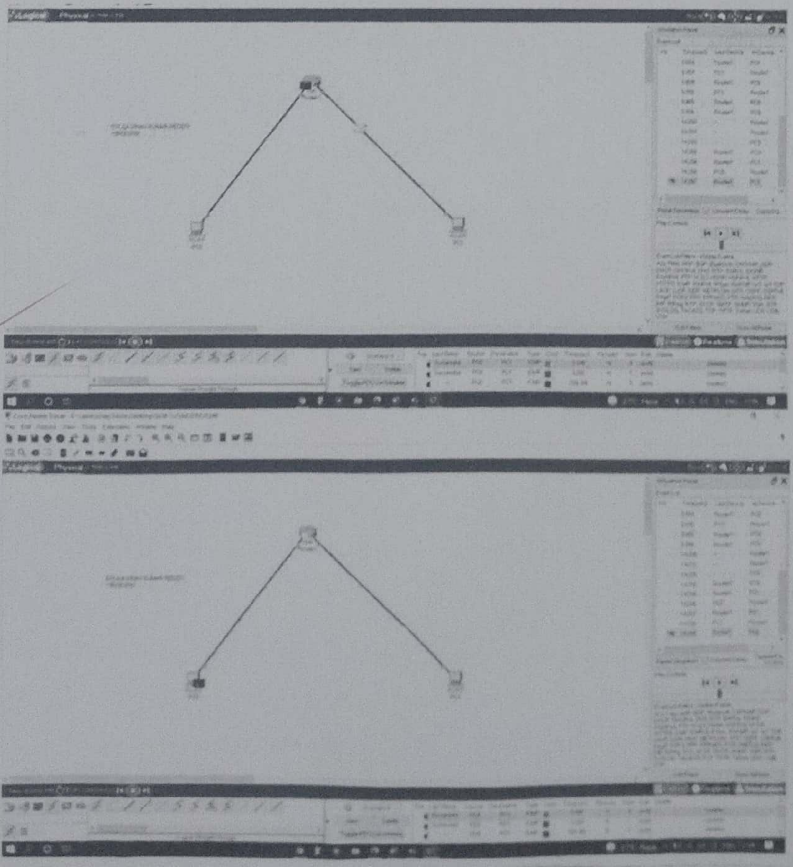
ii) Connect fastethernet 0 port of PC1 with fast ethernet 0/1 port of router1 using a copper straight through cable.

Device name	IP address (0/0)	Subnet mask.	IP address (0/1)	Subnet mask.
Router 1	192.168.10.1	255.255.255.0	192.168.20.1	255.255.255.0





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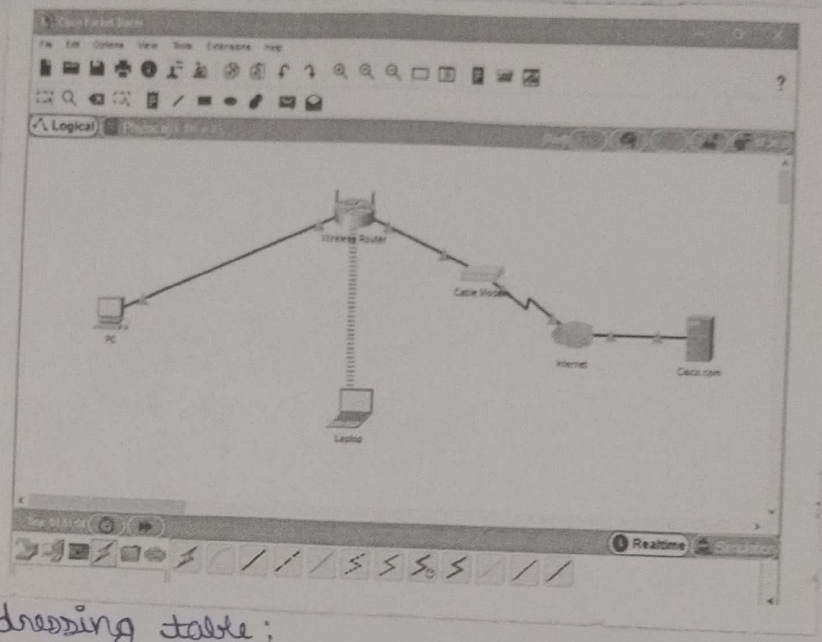
Ex.No:10b Design & Configure an internetwork using router, DHCP server.

Aim:

Design & Configure an internetwork using wireless router, DHCP server and internet cloud.



Addressing table:

Device	Interface	IP address	Subnet mask	default gateway
PC	ethernet 0.	DHCP		192,168.0.1
wireless router	Internet	DHCP		
laptop	wireless 0	DHCP.		

Objectives:

i) Build a simple network in the logical topology workspace.

ii) Configure the network devices.

iii) Test Connectivity between network devices.

iv) Save the file and close Packet Tracer.

i)

Step 1: Launch Packet Tracer.

Step 2: Build the topology.

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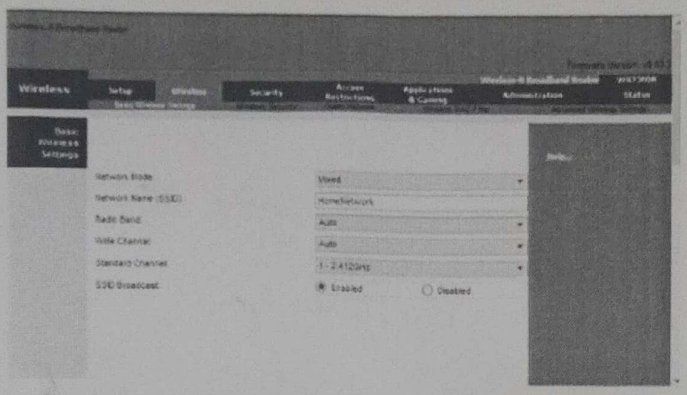
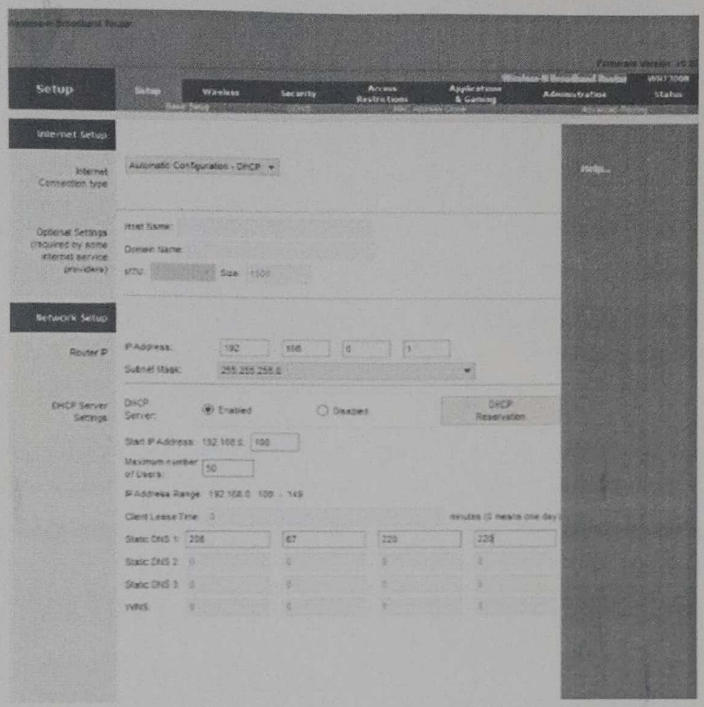
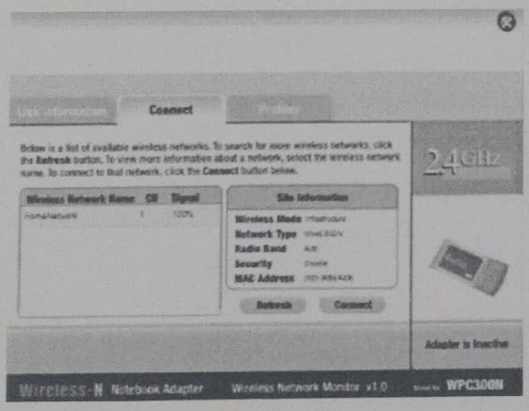
ii) Step 1: Configure the wireless router,

Step 2: Configure the laptop.







Step 3: Configure the PC,

Click on the PC icon on the packet tracer logical workspace and select the Desktop tab and then the IP configuration icon.

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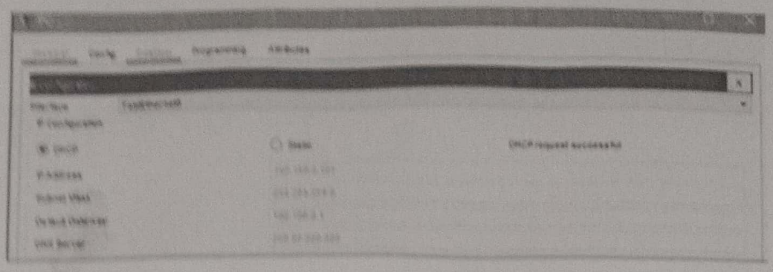
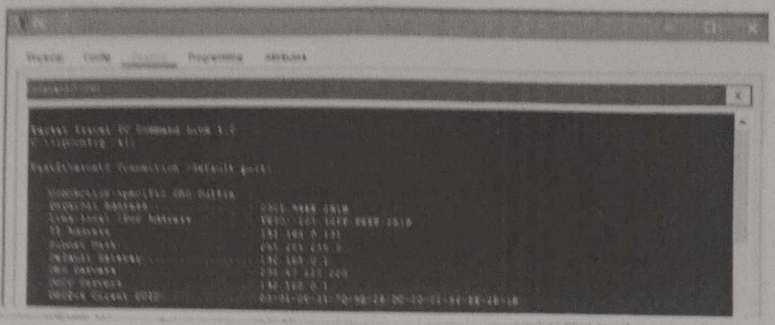
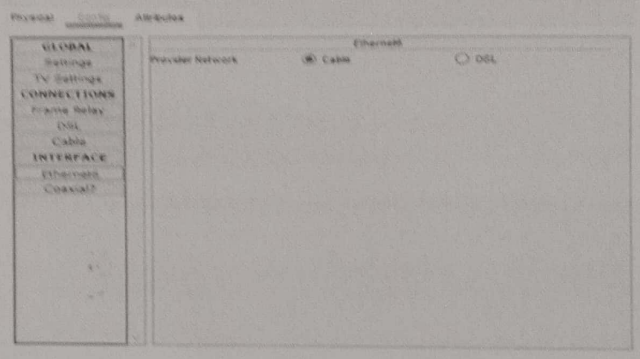
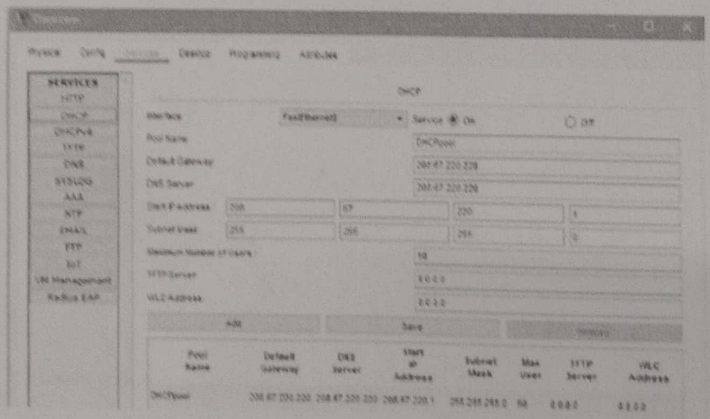


Click on the Command Prompt icon. Verify that the PC has received an IPv4 address by issuing the **ipconfig /all** command from the command prompt as shown in the figure. The PC should receive an IPv4 address in the 192.168.0.x range.



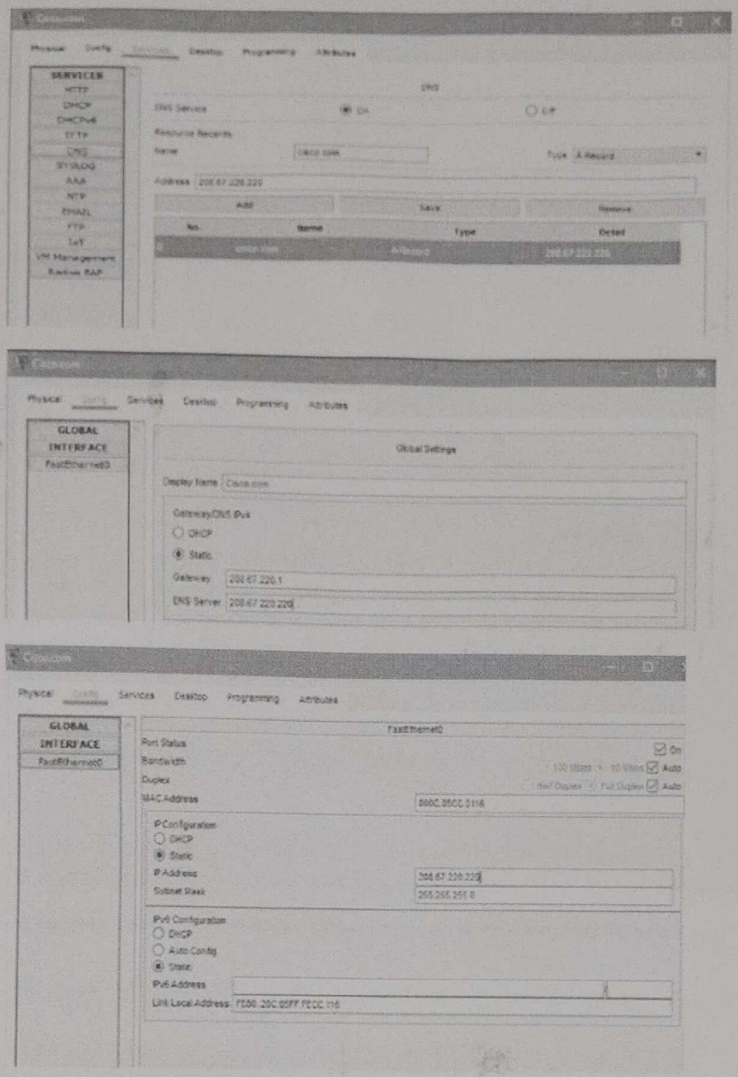
Step 4: Configure the Internet cloud.





Step 5: Configure the Cisco.com server.

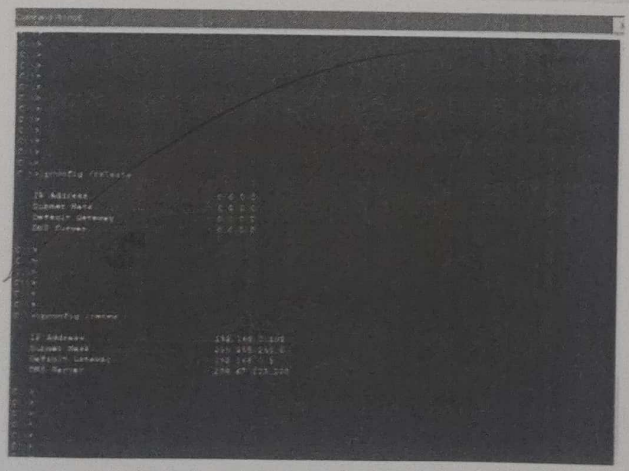
click add to add the pool.



(ii) Step 1:

Refresh the IPV4 settings on the PC.

a) Verify that the PC is receiving IPV4 configuration information from DHCP.



b) Test connectivity to the Cisco.com server from the PC.

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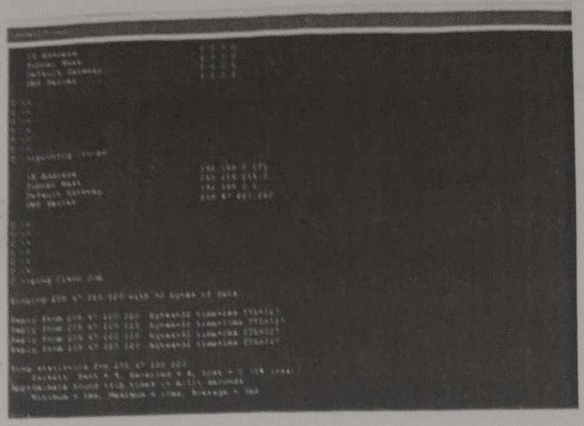

Student Observation;

1) write down the key features of configuring wireless router and DHCP server.

⇒ Wireless router;

* Provides wireless network connectivity.
* Configures SSID.

DHCP server;

* Assigns DNS server addresses.
* Avoids IP address conflicts.

2) What is the significance of DHCP server in inter-networking?

⇒ * Automatically assigns IP addresses.
* Reduces manual configuration.
* Simplifies network administration.
* Prevents duplicate IP address conflicts.

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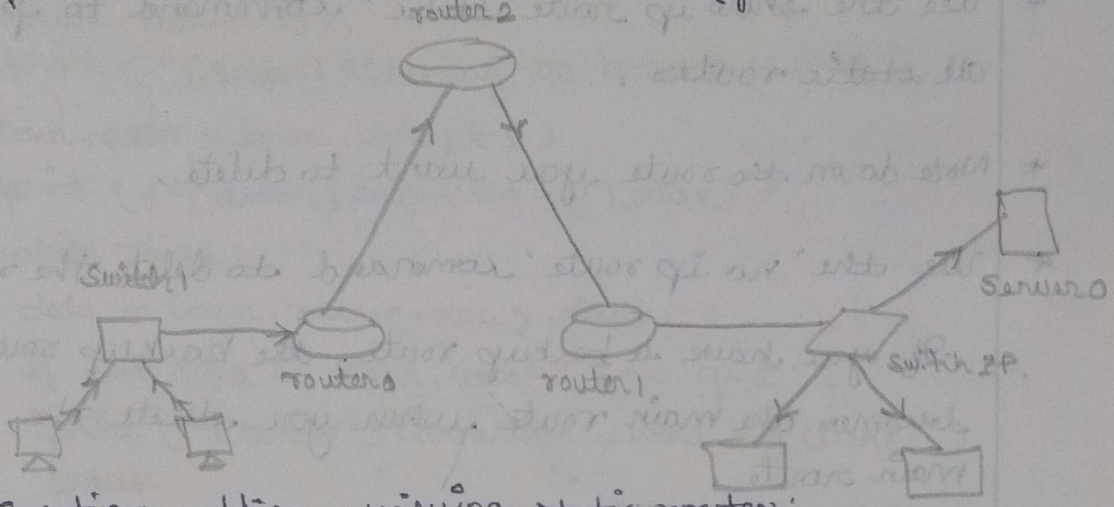
Ex:NO:11 Simulate Static routing Configuration using CISCO packet Tracer.

Aim:

Simulate static routing Configuration using CISCO packet tracer.

Setting up a lab:

Create a packet tracer lab as shown in the following image or download the following and load it on packet tracer with initial IP configuration.



Creating, adding, verifying static routes:

Routers automatically learn their Connected networks, we only need to add routes for the networks that are not available on the router's interfaces, for example, network 10.0.0.0/8, 20.0.0.0/8 and 40.0.0.0/8 are directly connected to router().

Router 1 requirements:

* Create two routes for network 10.0.0.0/8 and Configure the first route as the main route and the second route as a backup route.
* Verify the router adds only main routes to the routing table.

Router 2 requirements:

Create static routes for network 10.0.0.0/8 and network 30.0.0.0/8 and verify the router adds both

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routes to the routing table.

Verifying static routing:

On router(), we configured two routes for network 30.0.0.0/8. These routes are via router 1 and via router 2. We set the first route as the main route and the second route as the backup route.

Deleting a static route:

To delete a static route, use the following steps.

* Use the 'show ip route static' command to print all static routes.
* Note down the route you want to delete.
* Use the 'no ip route' command to delete the route.

If you have a backup route, the backup route becomes the main route when you delete the main route.

Router() (config)# router rip
router() (config-router)# network.
10.0.0.0
router() (config-router)# network 192.168.1.252.

⇒ router rip command tell router to enable the RIP routing protocol.

Network command allows us to specify the networks which we want to advertise.

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Ex.No:12a) 12. End - end communication at Transport layer.

Aim:

Implement echo client using TCP/UDP sockets.

Algorithm:

```
import socket
import threading
def start_server():
  server = socket.socket(socket.AF_INET, socket_stream)
server.bind(('127.0.0.1', 5000))
server.listen(1).
print("[server] listening on 127.0.0.1:5000")
conn, addr = server.accept()
print(f"[server] connected by [addr]")
while True:
  data = conn.recv(1024).decode()
  if not data or data.lower() == 'quit':
   print("[Server] Connection closed by client.")
   break
   print(f"[Server] received: {data}")
   conn.send(f"Server received: {data}".encode())
conn.close()
server.close().
if __name__=="__main__":
 server_thread = threading.thread(target = start_
              server, daemon = True)
 server_thread.start()
 start_client().
```

Output:

[server] listening on 127.0.0.1:5000 [client] connected to server

[server] connected by ('127.0.0.1', 53458) Enter message (or 'quit' to exit):

hello.

[Server] Received: hello.

[Server Response] Server received: hello.

Ex:No:12b) End-end communication at Transport layer.

Aim:

Implement chat client server using TCP/UDP packet sockets.

Algorithm:

```
import socket
import threading
def udp_server():
    s = socket.socket(socket.AF_INET, socket.SOCK_DGRAM
    s.bind(('127.0.0.1', 12000))
    print("[Server] running on 127.0.0.1:12000")
    while True:
    data, addr = s.recvfrom(1024)
    msg = data.decode()
    if msg.lower() == 'quit':
    print(f"[Server] client {addr} ended
                the connection.")
     break
     print(f"[Server] received from {addr}: {msg}")
     s.sendto(data, addr)
     s.close()
     print("[Server] Stopped.")
def udp_client():
    c = socket.socket(socket.AF_INET,
                socket.SOCK_DGRAM)
    server = ('127.0.0.1', 12000)
    while TRUE:
    msg = input("Enter message (or 'quit' to stop):")
    c.sendto(msg.encode(), server)
    if msg.lower() == 'quit':
        print("[CLIENT] closing connection.....")break
    data, _ = c.recvfrom(1024)
```

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```
s.close()
print("[Server] Stopped.")
def udp-client():
    c=socket.socket(socket.AF-INET, socket.SOCK_DGRAM)
    server=('127.0.0.1', 12000)
    while True:
        msg=input("Enter message (or 'quit' to stop):")
        c.sendto(msg.encode(), server)
        if msg.lower()=='quit':
            print("[client] closing connection....") break
        data,_=c.recvfrom(1024)
        print("Echo from Server:", data.decode())
    c.close()
    print("[client] Stopped.")
    if __name__=="__main__":
        threading.thread(target=udp-server,
                         daemon=True).start()
        udp-client().
```

Output:

Enter message (or 'quit' to stop): [Server] running on 127.0.0.1:12000 hi.

[Server] Received from ('127.0.0.1', 49670): hi

Echo from Server: hi

Enter message (or 'quit' to stop): hello

[Server] received from ('127.0.0.1', 49670): hello

Echo from server: hello.

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EX.NO:13 Implement your own ping program.

Aim:

Implement your own ping program.

Algorithm:

```
import os
import platform
import subprocess
def ping(host="google.com", count=4):
param="-n" if platform.system().lower()==
        "windows" else "-c"
Command=["ping", param, str(count), host]
try:
output=subprocess.check_output(Command, universal_
                                newlines=True)
print(output)
except Exception as e:
print(f"Ping failed: {e}")
if __name__=="__main__":
ping("8.8.8.8")
```

Output:

Pinging 8.8.8.8 with 32 bytes of data:

reply from 8.8.8.8: bytes = 32 time = 6ms TTL=119

reply from 8.8.8.8: bytes = 32 time = 9ms TTL=119.

Ping statistics for 8.8.8.8:

Packets: sent=4, received=4, lost=0 (0% loss)

Approximate round trip times in milli-seconds:

Minimum = 5ms, Maximum = 9ms, Average = 7ms.

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EX.NO:14 RAW sockets to implement packet sniffing.

Aim:

to determine the RAW sockets to implement packet sniffing.

Code:

```
from scapy.all import sniff
from scapy.layers.inet import IP,TCP,UDP,ICMP
def packet_callback(packet):
    if IP in packet:
        ip_layer = packet[IP]
        protocol = ip_layer.proto
        src_ip = ip_layer.src
        dst_ip = ip_layer.dst
        protocol_name = ""
        if protocol == 1:
            protocol_name = "ICMP"
        elif protocol == 6:
            protocol_name = "TCP"
        elif protocol == 17:
            protocol_name = "UDP"
        else:
            protocol_name = "Unknown Protocol".
        print(f"Protocol : {protocol_name}")
        print(f"Source IP: {src_ip}")
        print(f"Destination IP: {dst_ip}")
        print("-",*50)
    sniff(iface = 'Wi-Fi', prn = packet_callback,
                    filter = "ip", store = 0).
```

Output:

Protocol : TCP

Source IP : 57.155.141.17

Destination IP : 192.168.1.6

Protocol : UDP

Source IP : 192.168.1.6

Destination IP : 192.168.1.1

Completed

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